

COMP 1111 1. Midterm

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I. QUESTION (16 POINTS)

Write a program that create random passwords as follows: The number of digits is entered by the user (Must be between 3 and 5). Then a random password with the entered number of digits is created and displayed. If the number of digits is outside the interval, the program will display an error message.

```
Enter number of digits: 4
Password is: 7342
```

```
Enter number of digits: 7
Incorrect number of digits!
```

II. QUESTION (18 POINTS)

Write a program which takes a positive integer less than 16;

- prints "Divisible by 2", if the number is divisible by 2 (2, 4, 6, 8, 10, 12, 14).
- prints "Divisible by 3", if the number is divisible by 3 (3, 6, 9, 12, 15).
- prints "Prime", if the number is prime (2, 3, 5, 7, 11, 13)
- prints "Fibonacci", if the number is one of the fibonacci numbers (1, 2, 3, 5, 8, 13).

If one number satisfies more than one condition, all messages must be displayed. You must use switch-case construct in this problem.

```
Enter number: 2
Divisible by 2
Prime
Fibonacci
```

```
Enter number: 15
Divisible by 3
```

III. QUESTION (16 POINTS)

Three numbers are entered by the user (x, y, z) which are supposed to be the sides of a triangle, check if it is a valid triangle or not. Hint: the sum of any two sides should be greater than the other.

```
Enter three numbers
x: 3
y: 9
z: 7
Valid
```

```
Enter three numbers
x: 1.2
y: 10.2
z: 12.2
Invalid
```

IV. QUESTION (16 POINTS)

Write a program which takes three integers (x, y, z) from the user and prints the harmonic mean of those three integers. The harmonic mean of three numbers is calculated as

$$H = \frac{1}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$$

```
Enter three numbers:
x: 2
y: 3
z: 6
Harmonic Mean: 1
```

```
Enter three numbers:
x: 4
y: 4
z: 8
Harmonic Mean: 1.6
```

V. QUESTION (18 POINTS)

A number is called consecutive magical number if each digit differs by exactly the same amount. For example, each of the following four-digit integers are consecutive magical numbers: 1234 (difference is 1), 5432 (difference is -1), 3579 (difference is 2) and 9630 (difference is -3). Write a program that asks the user to enter a four-digit integer and determines if the given number is a consecutive magical number or not.

```
Enter a four-digit number: 4321
It is a consecutive magical number!
```

```
Enter a four-digit number: 1470
All digits are not consecutive and magical!
```

VI. QUESTION (16 POINTS)

Write a shopping discount calculator. First, ask user to enter the price of the item he wants to buy. Then, ask user to enter a number between 1 and 7 representing the day of the week he goes for shopping (1 represents Monday, 7 represents Sunday). The discount rate is

- 25% for odd weekdays (Monday, Wednesday)
- 20% for even weekdays (Tuesday, Thursday)
- 15% for weekends (Friday, Saturday, Sunday).

Your program will show the price of the item after the discount. You must use switch-case construct in this problem.

```
Enter price of the item: 280.99
Enter which day you go shopping: 6
You'll got 15% discount.
You'll pay 238.84
```